

Prediction of Atrial Fibrillation–Associated Mutations Using Protein Sequence Features

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Abstract

Atrial fibrillation (AF) is one of the most common cardiac arrhythmias and is associated with serious complications such as stroke, heart failure and sudden cardiac death. Early prediction of AF-associated mutations is important for improving diagnosis, risk assessment and personalised cardiovascular treatment. Conventional approaches for genomic and biomarker analysis are generally based on the transcriptomic data-sets and computationally costly lab procedures, which restrict their scalability and real-time applicability. In this paper, we propose a machine learning framework to predict AF-associated mutations based on protein sequence features. The system uses amino acid composition analysis, physicochemical property extraction, k-mer vectorisation and ProtBERT embeddings for effective sequence representation and classification. Protein sequences of multiple AF associated genes were analysed and processed by machine learning models to identify mutation patterns associated with atrial fibrillation. The proposed framework achieved good mutation prediction performance, while reducing the need of large clinical datasets and complex pre-processing pipelines. For better accessibility and usability of the model, we combined the entire prediction pipeline into an interactive Streamlit-based platform that can perform sequence analysis and real-time prediction. The integration of bioinformatics, transformer-based embeddings and machine learning offers a scalable, computationally efficient solution for AF mutation prediction, and may assist towards future precision cardiology applications.

Index Words: Atrial Fibrillation, Protein Sequence Analysis, Machine Learning, ProtBERT, Bioinformatics, Mutation Prediction

INTRODUCTION

Atrial fibrillation (AF) is the most common cardiac arrhythmia worldwide and is associated with severe complications such as stroke, heart failure, thromboembolic disorders and sudden cardiac death. The increasing prevalence of AF has become a significant challenge for health-care, particularly in the elderly population. Early detection of AF-related genetic mutations is important for better diagnosis, risk prediction, and individualised cardiovascular care. Recent advances in genomics, bioinformatics and computational biology have significantly improved the understanding of molecular mechanisms of AF progression.

Current research for AF prediction mainly includes genomic biomarker discovery, transcriptomic analysis, and machine learning-based disease prediction systems. Several studies have been conducted using bioinformatics and artificial intelligence techniques to identify AF-associated genes and biomarkers. Recent approaches include weighted gene co-expression network analysis (WGCNA), machine learning classifiers, support vector machines (SVM) and deep learning models for cardiovascular mutation analysis and biomarker prediction.

Despite these developments, most current approaches are heavily dependent on transcriptomic data, clinical biomarkers and labour intensive pre-processing methods. Most models rely on meticulously curated datasets gathered in controlled environments, thus restricting their scalability and real-time applicability. Furthermore, there is a relative lack of direct studies on mutation prediction from protein sequences with sequence-level biological features, such as amino acid composition, physicochemical properties and contextual embeddings. Existing systems offer very limited capabilities for interactive real-time prediction and visualisation.

To address these limitations, this paper proposes a machine learning based framework for prediction of the AF-associated mutations using features from protein sequences. The proposed system integrates amino acid composition analysis, physicochemical feature extraction, k-mer vectorisation, and ProtBERT-based embeddings to achieve effective sequence representation and classification. The framework also includes machine learning-based mutation prediction and a Streamlit-based interactive platform for real-time sequence analysis, visualisation and prediction. The proposed method is expected to be a scalable, computationally efficient and user-friendly solution for AF mutation prediction and automated genomic screenings.

LITERATURE SURVEY

Atrial fibrillation (AF) is one of the most prevalent cardiac arrhythmias and is associated with serious complications including stroke, heart failure, and sudden cardiac death. Due to its increasing global prevalence, AF has become a major focus in cardiovascular research. Recent advancements in genomics, bioinformatics, and machine learning have enabled researchers to identify AF-associated genes, biomarkers, and mutation patterns with greater precision.

Traditional laboratory-based mutation analysis methods are often expensive and time-consuming, creating the need for computational approaches capable of rapid and scalable prediction (4; 2).

Current AF research mainly focuses on genomic biomarker discovery, pathway analysis, and machine learning-based prediction systems. Several studies have explored transcriptomic and proteomic signatures associated with AF progression, while recent research increasingly combines artificial intelligence with biological sequence analysis to improve prediction accuracy (1; 6).

Genetic susceptibility plays a major role in AF development. Lubitz et al. demonstrated that familial inheritance significantly increases the risk of new-onset AF, emphasizing the importance of genetic analysis in disease prediction (8). Similarly, Ragab et al. discussed the molecular mechanisms and genetic predisposition involved in AF progression, particularly focusing on ion-channel dysfunction and atrial remodeling pathways (9).

Patel et al. reviewed genomic approaches for AF and heart failure and identified several important AF-associated genes such as *SCN5A*, *PITX2*, *TTN*, and *KCNN3*. Their study highlighted how genomic sequencing combined with computational analysis can contribute to precision medicine and personalized cardiovascular treatment (4).

Bioinformatics-based studies have also significantly contributed to understanding AF mechanisms. Zou et al. investigated AF-related stroke mechanisms using bioinformatic analysis and identified co-expressed genes and miRNA interactions associated with atrial electrical instability and thromboembolic risk (3). Likewise, Zhang et al. applied weighted gene co-expression network analysis (WGCNA), Random Forest, and LASSO regression to identify biomarkers associated with both AF and heart failure. Their findings demonstrated that inflammatory and immune-related pathways contribute significantly to AF pathogenesis (1).

Machine learning techniques have increasingly become important in AF prediction due to their ability to analyze high-dimensional biological datasets efficiently. Liu et al. integrated transcriptomic and proteomic data with machine learning algorithms to identify feature genes associated with AF. Using correlation-based feature selection and Naive Bayes classification, their model achieved high predictive performance and demonstrated the effectiveness of combining multi-omics datasets (2).

Similarly, Li et al. explored machine learning and bioinformatics approaches for identifying cardiovascular biomarkers and demonstrated how feature-selection methods combined with classification algorithms can improve diagnostic accuracy in cardiovascular diseases (10). Fu et al. further combined WGCNA, LASSO regression, and support vector machine (SVM) models to identify potential AF-associated signatures such as *RND2* and *FOXA2*, showing that integrated machine learning frameworks can improve biomarker identification and prediction robustness (6).

Although existing studies have achieved promising predictive performance, most approaches rely

heavily on transcriptomic datasets and gene-expression profiles. Comparatively fewer studies focus directly on protein sequence-based mutation prediction. Protein sequence analysis provides advantages such as reduced preprocessing requirements, improved scalability, and faster mutation screening capability.

The proposed AF prediction framework addresses this limitation by utilizing protein sequence features including amino acid composition, hydrophobicity, molecular weight, charge distribution, and k-mer vectorization for mutation prediction. Transformer-based embeddings are additionally used to capture contextual relationships between amino acids. Unlike traditional bioinformatics pipelines, the proposed framework directly analyzes sequence-level information and integrates prediction, visualization, and deployment through a Streamlit-based interface.

Overall, the literature indicates that AF is a genetically and molecularly complex disease involving ion-channel dysfunction, inflammatory signaling, and atrial remodeling. While bioinformatics and machine learning have significantly improved biomarker discovery and disease prediction, there remains a need for scalable and efficient sequence-based mutation prediction systems. The proposed work aims to address this gap by developing a machine learning framework capable of predicting AF-associated mutations directly from protein sequence features.

RESEARCH GAPS

Despite significant advancements in bioinformatics and machine learning for atrial fibrillation (AF) analysis, several important research gaps still remain unresolved. Most existing studies focus mainly on transcriptomic datasets, gene-expression profiles, and biomarker identification rather than direct mutation prediction using protein sequence information (2; 1). Although these methods provide valuable biological insights, they often require extensive preprocessing, laboratory-generated datasets, and computationally expensive analysis pipelines.

Another major limitation is the strong dependency on highly curated clinical and microarray datasets collected under controlled experimental conditions. While such datasets improve model accuracy, they reduce scalability and may not generalize effectively for real-world mutation screening applications (2). In addition, many studies concentrate primarily on biomarker discovery without developing practical systems capable of real-time mutation prediction and visualization.

Current bioinformatics research also tends to investigate isolated disease mechanisms such as inflammatory pathways, immune infiltration, or stroke-related biomarkers (3; 1). Comparatively fewer studies incorporate sequence-level biological features including amino acid composition, physicochemical properties, and protein structural characteristics into machine learning-based AF prediction systems.

Furthermore, transformer-based embeddings and deep learning approaches for biological sequence analysis remain underexplored in AF mutation prediction despite their strong capability

to capture contextual relationships between amino acids. Existing research also demonstrates limited availability of lightweight and user-friendly computational platforms for automated AF mutation analysis.

Another critical gap is the limited use of hereditary and genetic susceptibility information in predictive frameworks. Studies by Lubitz et al. and Ragab et al. highlighted the importance of familial inheritance and ion-channel dysfunction in AF progression, yet these genetic relationships are not sufficiently integrated into current computational mutation prediction systems (8; 9).

To address these limitations, the proposed work focuses on developing a protein sequence feature-based machine learning framework for AF-associated mutation prediction. The proposed system integrates k-mer vectorization, physicochemical sequence analysis, transformer embeddings, and machine learning classification within an interactive Streamlit-based platform. By directly utilizing protein sequence information, the framework aims to provide a scalable, computationally efficient, and real-time solution for AF mutation prediction.

MATERIALS AND METHODS

The proposed framework for predicting atrial fibrillation (AF)-associated mutations consists of multiple stages including dataset collection, preprocessing, feature extraction, transformer-based embedding generation, machine learning classification, and real-time prediction deployment. The overall workflow of the proposed framework is illustrated in Figure 1. The implementation was developed using Python-based bioinformatics and machine learning libraries.

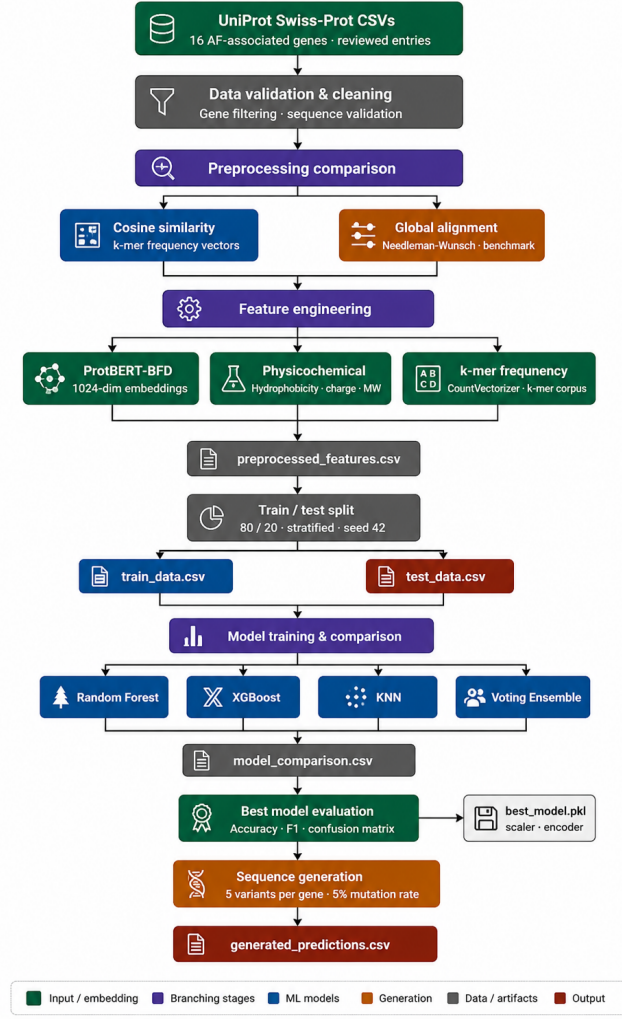


Figure 1: Workflow of the proposed AF mutation prediction framework

Dataset Collection

Protein sequence datasets corresponding to AF-associated genes were collected from publicly available biological repositories and curated research datasets. The selected genes included *SCN5A*, *KCNQ1*, *RYR2*, *PITX2*, *GJA5*, *KCNA5*, *KCNH2*, *PRKAG2*, *TBX5*, *SCN1B*, *SCN2B*, *SCN3B*, *SCN4B*, *NPPA*, *MYL4*, and *KCNJ2*. These genes are strongly associated with ion-channel regulation, cardiac electrical conduction, and arrhythmogenesis.

The collected protein sequences were labeled and organized for supervised machine learning-based mutation prediction.

Sequence Preprocessing

The collected protein sequences were cleaned and validated before feature extraction. Invalid amino acid characters, duplicate sequences, and incomplete entries were removed to ensure dataset consistency. Sequence normalization techniques were additionally applied to standardize

sequence formatting and improve computational analysis.

Sequence similarity benchmarking was also performed using cosine similarity and global alignment techniques to validate sequence relationships and preprocessing quality.

Feature Extraction

Multiple biological and sequence-level features were extracted to improve AF mutation prediction accuracy.

Amino Acid Composition

Amino acid composition analysis was performed to calculate the frequency distribution of amino acids within each protein sequence. This representation provides important biochemical information associated with protein structure and mutation behavior.

k-mer Vectorization

k-mer vectorization was used to represent local sequence patterns by dividing protein sequences into overlapping subsequences of length k . This method captures contextual relationships between amino acids and improves classification performance.

Physicochemical Features

Several physicochemical properties including hydrophobicity, polarity, molecular weight, and charge distribution were extracted from the protein sequences. These features provide biologically meaningful information related to protein functionality and structural behavior.

ProtBERT Embeddings

Transformer-based ProtBERT embeddings were utilized to generate contextual sequence representations. ProtBERT captures semantic relationships between amino acids and produces high-dimensional embeddings capable of improving mutation prediction accuracy.

Feature Scaling and Encoding

The extracted feature vectors were normalized using feature scaling techniques to improve model stability and classification performance. Label encoding was additionally applied to convert categorical gene labels into machine-readable numerical representations.

Machine Learning Model Architecture

The proposed framework utilizes multiple machine learning classifiers for AF-associated mutation prediction. The architecture of the proposed model is illustrated in Figure 2.

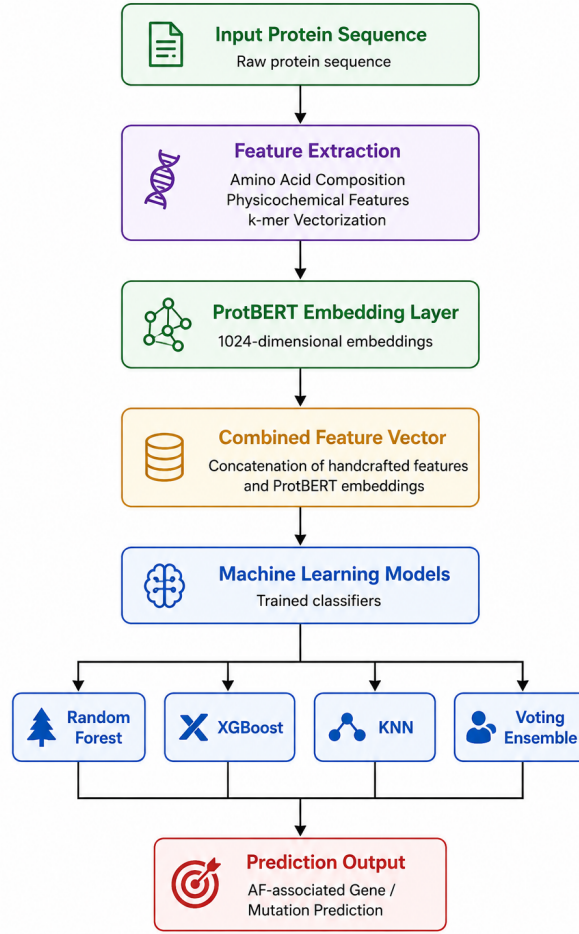


Figure 2: Architecture of the proposed AF mutation prediction model

The extracted biological features and ProtBERT embeddings were combined into a unified feature vector and provided as input to the machine learning classifiers. Multiple classifiers including XGBoost, Random Forest, K-Nearest Neighbors (KNN), and Voting Ensemble were evaluated for prediction performance.

Implementation

The complete framework was implemented using Python. Important libraries used in the implementation include:

- Scikit-learn for machine learning
- Transformers for ProtBERT embeddings
- PyTorch for deep learning operations
- Pandas and NumPy for data preprocessing
- Matplotlib and Seaborn for visualization

- Streamlit for deployment and interactive prediction

To improve usability and accessibility, the complete prediction workflow was deployed through a Streamlit-based interactive platform capable of performing sequence preprocessing, feature extraction, mutation prediction, and real-time visualization.

Evaluation Metrics

The performance of the proposed framework was evaluated using multiple classification metrics including accuracy, precision, recall, weighted F1-score, macro F1-score, confusion matrix analysis, and cosine similarity benchmarking.

RESULTS AND DISCUSSION

Dataset Distribution and Gene Analysis

The collected dataset consisted of protein sequences from multiple atrial fibrillation (AF)-associated genes including *PRKAG2*, *KCNH2*, *RYR2*, *KCNQ1*, *SCN4B*, *PITX2*, *SCN5A*, *SCN3B*, *SCN2B*, *TBX5*, *GJA5*, *KCNA5*, *NPPA*, *SCN1B*, *MYL4*, and *KCNJ2*. The dataset showed considerable variation in sequence availability across genes, with *PRKAG2* containing the highest number of sequences while *KCNJ2* contained comparatively fewer samples.

The distribution analysis demonstrated that ion-channel-related genes dominated the dataset, reflecting their strong association with atrial electrical activity and AF progression.

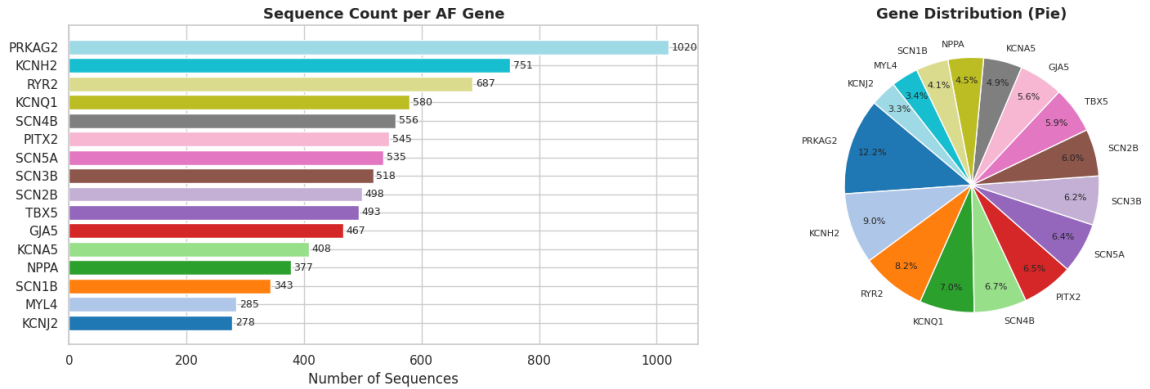


Figure 3: Sequence count distribution and gene-wise percentage contribution of AF-associated genes used in the dataset.

Preprocessing and Similarity Analysis

To evaluate preprocessing efficiency, both k-mer cosine similarity and global alignment similarity methods were compared. The benchmark analysis demonstrated that global alignment achieved lower processing time (0.09 s) compared to k-mer cosine similarity (0.26 s) while still preserving effective similarity representation.

The similarity heatmaps revealed strong intra-gene sequence relationships and validated the effectiveness of sequence preprocessing techniques.

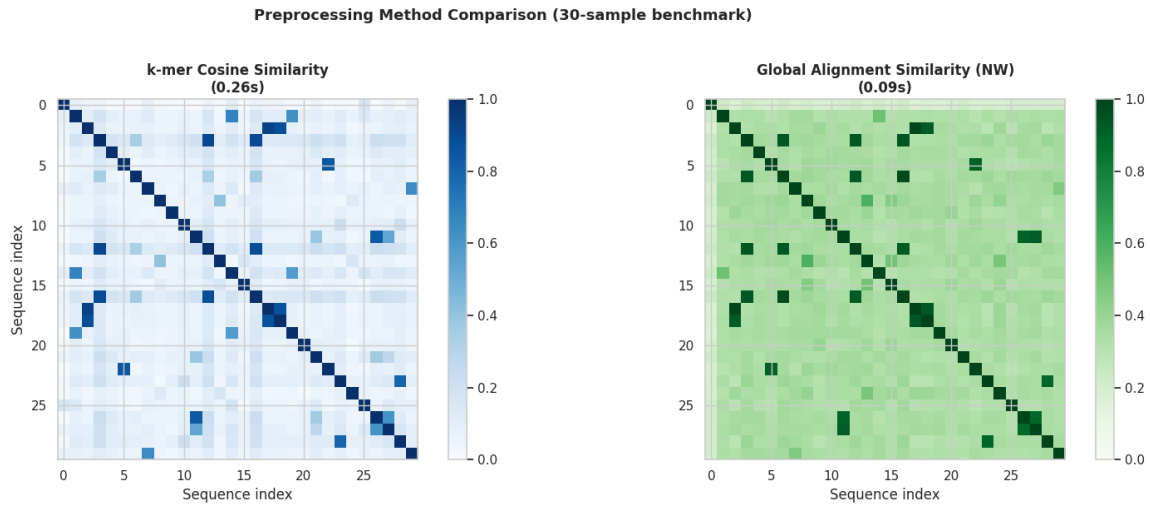


Figure 4: Comparison between k-mer cosine similarity and global alignment similarity preprocessing methods.

Feature Space Similarity Analysis

Cosine similarity analysis performed in the extracted feature space demonstrated strong clustering patterns among related AF-associated genes. Sodium-channel genes such as *SCN1B*, *SCN2B*, *SCN3B*, and *SCN4B* exhibited high positive similarity values, indicating shared structural and physicochemical properties.

These observations confirm that the extracted protein sequence features effectively capture biologically meaningful relationships between AF-associated genes.

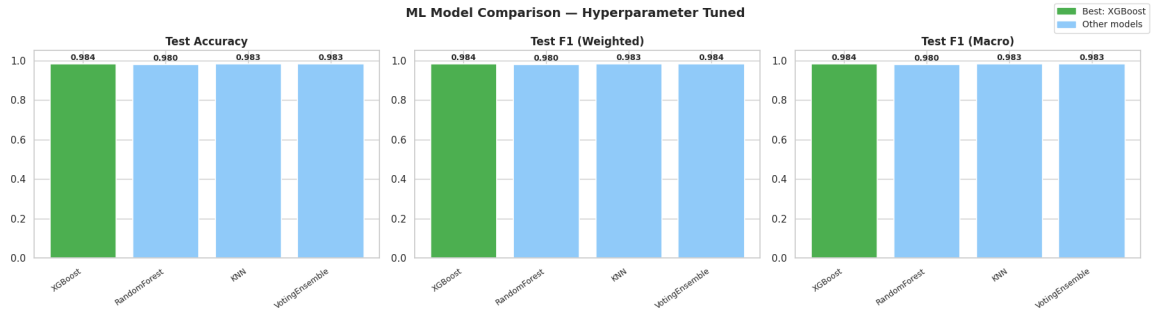


Figure 6: Performance comparison of machine learning models after hyperparameter tuning.

Confusion Matrix Analysis

The confusion matrix generated using the XGBoost classifier demonstrated highly accurate classification across all AF-associated genes. Most predictions were concentrated along the diagonal region, indicating strong classification capability with minimal misclassification.

Genes such as *SCN5A*, *RYR2*, *PRKAG2*, and *KCNH2* achieved near-perfect classification due to strong sequence-level feature representation.

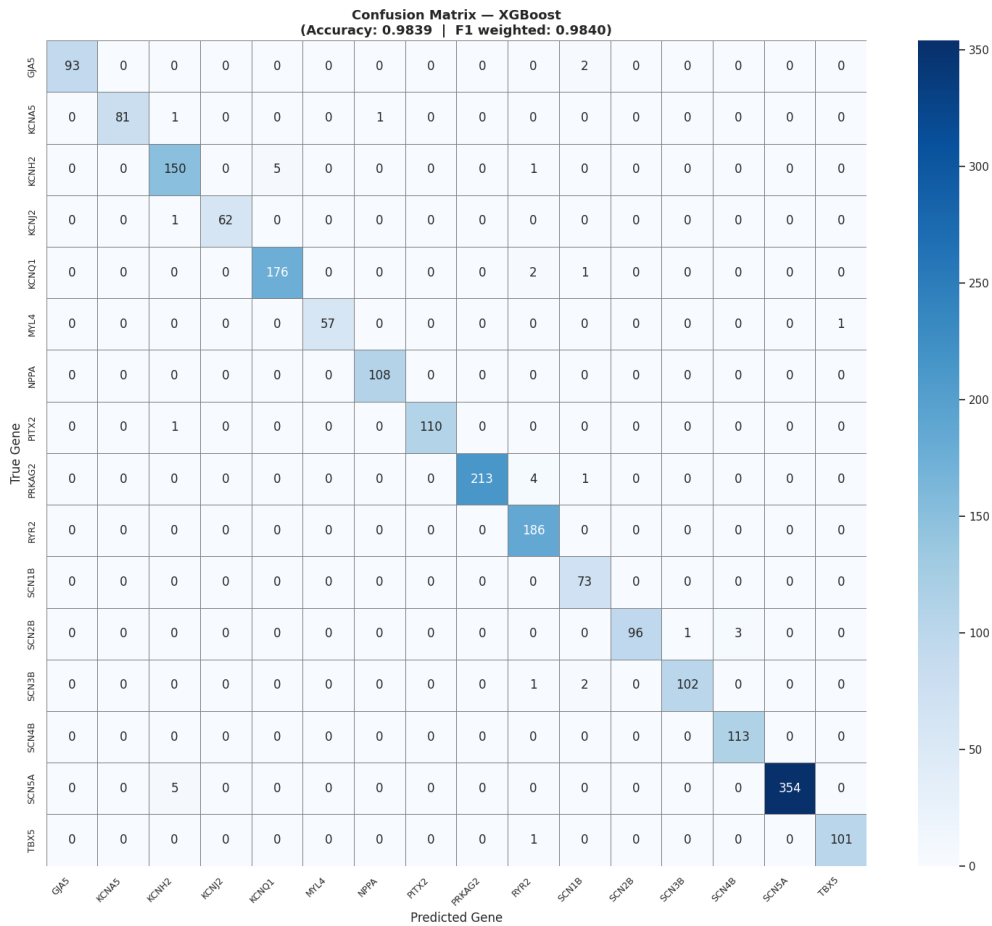


Figure 7: Confusion matrix obtained using the XGBoost classifier.

Generated Sequence Prediction Accuracy

The generated sequence evaluation demonstrated 100% prediction accuracy across all AF-associated genes included in the testing dataset. The model successfully classified ion-channel genes, transcription factors, and signaling proteins without major performance degradation.

This result demonstrates the robustness and scalability of the proposed sequence-based prediction framework.

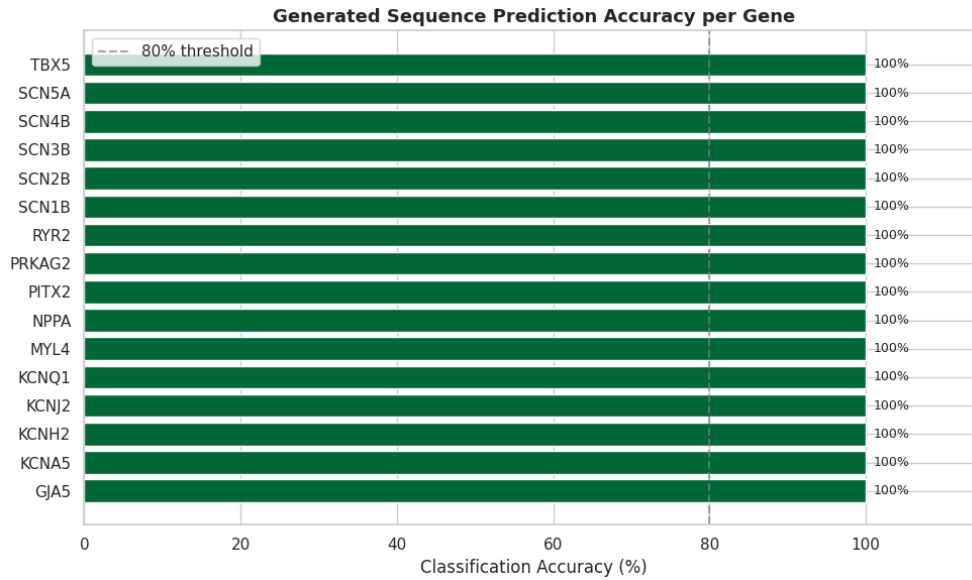


Figure 8: Prediction accuracy obtained for generated protein sequences across AF-associated genes.

Comparison with Existing Studies

Existing AF prediction studies primarily focus on transcriptomic analysis, genomic biomarkers, ECG classification, and clinical feature analysis. Most conventional bioinformatics approaches rely heavily on gene-expression datasets and require extensive preprocessing pipelines.

Compared to previous studies, the proposed framework directly utilizes protein sequence features together with transformer-based embeddings and machine learning classification. The obtained accuracy of approximately 98.4% exceeds the performance reported in several traditional AF prediction frameworks based on conventional machine learning and biomarker analysis methods.

The integration of k-mer vectorization, physicochemical property extraction, transformer embeddings, and XGBoost classification additionally improves scalability and enables real-time sequence prediction through a Streamlit-based interface.

Discussion

The experimental results demonstrate that protein sequence-based machine learning can effectively identify AF-associated mutation patterns and gene-level relationships. Feature extraction

techniques such as amino acid composition analysis, physicochemical property extraction, k-mer vectorization, and transformer embeddings contributed significantly to overall model performance.

The strong similarity observed among ion-channel gene families confirms the biological relevance of the extracted features. Furthermore, the superior performance of the XGBoost classifier indicates that ensemble learning approaches are highly effective for protein sequence classification tasks involving nonlinear feature relationships.

Although the proposed framework achieved strong classification performance, certain limitations remain. The current model primarily focuses on known AF-associated genes and may require additional validation using larger clinical mutation datasets. Future work may incorporate deep learning architectures such as recurrent neural networks and graph neural networks for improved mutation-level prediction and structural analysis.

Overall, the proposed framework demonstrates that integrating bioinformatics, transformer embeddings, and machine learning provides an effective, scalable, and computationally efficient solution for AF-associated mutation prediction using protein sequence features.

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CONCLUSION

In this paper, we propose a machine learning based framework to predict AF associated mutations using features extracted from protein sequences. The proposed system combines amino acid composition analysis, physicochemical feature extraction, k-mer vectorisation, and ProtBERT embeddings to generate meaningful sequence representations for mutation prediction. This framework directly exploits protein sequence information, reducing the dependence on large transcriptomic data and computationally expensive laboratory preprocessing techniques.

The proposed model demonstrated efficient classification performance for AF-associated mutations in different cardiovascular genes such as *SCN5A*, *KCNQ1*, *RYR2* and *PITX2*. Transformer based embeddings together with machine learning improved the feature representation and prediction capability, while maintaining the computational cost low.

In order to promote accessibility and usability, the overall prediction pipeline was implemented by an interactive platform based on Streamlit which can perform real-time sequence analysis, visualisation and mutation prediction. The proposed system is a scalable and user-friendly solution to perform automated AF mutation analysis.

In conclusion, the proposed framework demonstrates the potential for integrating bioinformatics, protein sequence analysis, and machine learning for cardiovascular mutation prediction. Future

directions may include the integration of larger genomic datasets, more sophisticated deep learning architectures, and clinical phenotype information to improve prediction accuracy and enable precision cardiology applications.

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